CSC 350 Assignment 3
Spring 2025

Due: Tuesday, February 11 (Before Class)

General Instructions

- Scan or compile snapshots of your submission into a PDF and upload it to Canvas.
- Assignments are **due before the class** on the scheduled due date; generally, late assignments will not be accepted.
- Unless specified, **justify your answers always**. If justifications are missing or do not show reasonably how your answers are derived, 0 points will be assigned regardless of the correctness of the answers.
- Write **neatly** in the event you write your answers on paper. **Especially, for table completion questions, if your answers are not given neatly in well-drawn tables, the answers won't be graded**.
- **This assignment must be done by yourself without any unauthorized assistance**. Write your full name and numeric university ID on the front page of your submission, along with the following pledge statement.

I pledge my honor that all parts of this assignment were done by me individually and without any unauthorized assistance or collaboration with others.

Q1. Well-Formed Formulas

Explain why the following expression is a wff. Also, specify what each of the letters x, y, p, q, f designates (variable, constant, predicate, or function). Hint: See terms, atomic formula, and the definition of a wff in section 7.1.

(a) $\exists x \, \forall y \, (p(y) \rightarrow q(f(x), y))$

Q2. Scope, Bound, Free

For the following wff, label each occurence of the variables as either free or bound:

(a) $\forall y \, q(y) \wedge q(x) \wedge \neg \, \exists x \, p(x, y)$

Q3. Propositional Logic

Write down the propositional logic expression denoted by the following expression, where the variables take values in the domain $\{a, b\}$.

(a) $\forall y \, \exists x \, (p(x, y) \rightarrow q(y))$

Q4. Quantified Predicate Logic

Write down a quantified predicate logic expression over some domain to denote each of the following expressions. Specify the domain too.

(a) $q(a) \vee q(b)$

(b) $p(x, a) \wedge p(x, b)$

(c) $(p(a, a) \wedge p(a, b)) \vee (p(b, a) \wedge p(b, b))$

Q5. Interpretations

Given the wff $W = \exists x \, p(x) \rightarrow \forall y \, p(y)$, list ALL possible interpretations of W over the domain $\{a, b\}$. Also, give the truth value of W over each of the interpretations.

Hint: For each interpretation consider different cases where $p(a)$ and $p(b)$ are either true or false.

Q6. Models

Find a model for each of the following wffs:

(a) $\exists x \, p(x) \rightarrow \forall x \, p(x)$

(b) $\forall x \, (p(x, f(x)) \rightarrow p(x, y))$

Q7. Countermodels

Find a countermodel for each of the following wffs:

(a) $\exists x \, p(x) \rightarrow \forall x \, p(x)$

(b) $\forall x \, (p(x, f(x)) \rightarrow p(x, y))$

Q8. Validation

(a) Justify (argue, prove, ...) that the following formula is valid.

$$\forall x\, A(x) \vee \forall x\, B(x) \rightarrow \forall x\, (A(x) \vee B(x))$$

(b) Show that the following formula is not valid.
(Hint: Write an interpretation that makes it false.)

$$\forall x\, (A(x) \vee B(x)) \rightarrow \forall x\, A(x) \vee \forall x\, B(x)$$

Q9. Formal Quantified Expression

(a) Let $B(x)$ mean x is a bird,
let $W(x)$ mean x is a worm, and
let $E(x, y)$ mean x eats y.

Find an English sentence to describe the following expression:

$$\forall x\, \forall y\, (E(x, y) \rightarrow B(x) \wedge W(y))$$

(b) Let $e(x, y)$ mean that $x = y$,
let $p(x, y)$ mean that $x < y$, and
let $d(x, y)$ mean that x divides y.

For the following statement about the natural numbers, find a formal quantified expression.

"Any two nonzero natural numbers have a common divisor."